

What makes a chord progression with a melody liked — findings

Generated by the analysis script `discover_melody.py`, with a fixed random seed so the numbers reproduce exactly. Each rated item is 4 chords (16 notes) with a 10-note melody on top.

Statistical methods used — and why

- 1. Ratings treated as ordered, not as categories.** Dislike, not sure, and like are scored -1 , 0 , $+1$ and treated as an *ordered* scale. *Why*: it lets us ask the stronger question "does this pattern move the rating steadily in one direction?" rather than the weaker "does this pattern appear more in one bucket?" We verify the ordering is real rather than assuming it.
- 2. Features are pure arithmetic on the note numbers.** Every quantity (intervals, spreads, movement, clash counts, melodic shape) is computed by adding and subtracting the note values. *Why*: no music-theory used.
- 3. Split into a discovery half and a held-out half (60/40).** Patterns are *found* on the discovery half only, then *tested* on the held-out half, which was never examined during the search. *Why*: A pattern found and scored on the same data is almost guaranteed to look significant. Confirming on untouched data means a surviving pattern had to predict something it was not fitted to.
- 4. Rank correlation for the initial screen.** Each feature is ranked against the ordered rating (Spearman rank correlation). *Why*: rank-based methods assume only that the relationship is consistently increasing or decreasing.
- 5. Subgroup discovery to find pockets, not just trends.** A search over threshold conditions and their pairs, scored by how far a subgroup's average rating departs from the overall average, weighted by how many items it covers. *Why*: a feature can be useless overall yet decisive within a region (for example only when the melody is already high). Weighting by size stops the search from chasing tiny freak subgroups of three or four items.
- 6. Gradient-boosted trees as a cross-check.** A non-linear model is fitted and its feature importances inspected. *Why*: it catches curved relationships and interactions that a rank correlation would miss, so promising features are not overlooked. It is used only to *nominate* candidates.

7. Permutation tests for confirmation. On the held-out half, ratings are randomly reshuffled thousands of times to build the distribution of effects expected from chance alone; the real effect is compared to it. *Why*: this makes no assumption whatsoever about the shape of the data. It answers exactly the right question: "how often would shuffled ratings produce a gap this large?"

8. Effect size reported alongside significance (Cliff's delta). This measures how often items matching a pattern are rated above items that do not, on a scale from -1 to $+1$, where 0 means no difference. *Why*: a small p-value only says an effect is unlikely to be zero; it does not say the effect is *large*. Reporting both prevents a statistically detectable but musically trivial tendency from being oversold.

9. False-discovery-rate correction across all tests (Benjamini–Hochberg). The p-values from all confirmation tests are corrected together. *Why*: testing many patterns at the usual 5% threshold means roughly 1 in 20 pure-noise patterns passes. This correction controls the share of false positives among the patterns we declare real, which is the appropriate goal for a discovery study.

10. Bootstrap confidence intervals for stability. Each surviving pattern is re-measured on a thousand resamples of the held-out data. *Why*: a pattern that survives testing but whose interval straddles zero is riding on a handful of influential items. Only patterns whose effect keeps the same sign across resamples are reported as confirmed.

11. A noise ceiling from near-identical progressions. For each progression we check how often its most musically similar neighbours received a *different* rating. *Why*: with one rater and no repeated presentations, we cannot measure reliability directly. If near-identical items are rated inconsistently, that inconsistency caps how well *any* method could possibly do — and stops us blaming a weak result on the analysis when the ceiling is the real constraint.

12. A whole-model signal check against a shuffled baseline. A classifier is trained with cross-validation to separate liked from not-liked, and its score is compared against the same classifier trained on randomly shuffled ratings. *Why*: this is the honest global question — is there enough signal here to predict a rating at all? It is reported even when the answer is unflattering, because individual patterns can be real while still not adding up to a usable predictor.

How to read the results tables. *Effect size* runs from -1 to $+1$: positive means the pattern is rated higher than the rest, negative means lower, and roughly 0.1 is small, 0.3 moderate. The *95% interval* is the plausible range for that effect; if it crosses zero the pattern is unstable. The *p-value* is the chance of seeing an effect this large if the pattern were pure noise. *Number matching* is how many progressions the pattern actually covers.

Summary of what was found

- **248 candidate patterns** were screened (listed in the companion file `candidate_patterns.csv`).
- **Direction of the signal:** every surviving pattern points the same way: a **higher, rising melody** (its middle and later notes sitting high) combined with an **upward drift in the second-lowest voice** leans toward **liked**; a low or sagging melody leans toward **disliked**. Once the melody is included, melody height rather than chord dissonance is the dominant driver.
- **Where 'not sure' falls:** on a single dislike-to-like scale, dislike sits at -1.61, not sure at -1.13 and like at -0.31 → 'not sure' does fall between the other two, so treating the ratings as an ordered scale is justified.
- Effect sizes are **small** throughout (roughly 0.1 to 0.2 on a -1 to +1 scale).

Strongest confirmed pattern

the net rise or fall of the second-lowest voice is high (> 2) AND the melody's 6th note (pitch height) is high (> 68) → LIKE

Effect size +0.25 (95% interval [+0.11, +0.39]), p-value 0.0007, matching 65 progressions.

Confirmed patterns (significant after correction, with a stable effect)

pattern	direction	number matching	effect size	95% interval	p-value
the net rise or fall of the second-lowest voice is high (> 2) AND the melody's 6th note (pitch height) is high (> 68)	LIKE	65	+0.25	[+0.11, +0.39]	0.0007
the melody's 8th note (pitch height) is high (> 68) AND the net rise or fall of the second-lowest voice is high (> 2)	LIKE	66	+0.23	[+0.09, +0.37]	0.0010
the net rise or fall of the second-lowest voice is high (> 2) AND the	LIKE	69	+0.22	[+0.07, +0.36]	0.0023

pattern	direction	number matching	effect size	95% interval	p-value
melody's 7th note (pitch height) is high (> 68)					
the melody's 9th note (pitch height) is high (> 70)	LIKE	140	+0.17	[+0.05, +0.28]	0.0037
the melody's 8th note (pitch height) is high (> 70)	LIKE	139	+0.16	[+0.06, +0.27]	0.0050
the melody's 7th note (pitch height) is high (> 70)	LIKE	138	+0.14	[+0.03, +0.25]	0.0127

All confirmation tests (confirmed and failed)

Every candidate carried from the discovery half into held-out permutation testing, ordered by p-value.  survived the false-discovery-rate correction with a stable effect;  failed.

pattern	direction	effect size	p-value	verdict
the net rise or fall of the second-lowest voice is high (> 2) AND the melody's 6th note (pitch height) is high (> 68)	LIKE	+0.25	0.0007	
the melody's 8th note (pitch height) is high (> 68) AND the net rise or fall of the second-lowest voice is high (> 2)	LIKE	+0.23	0.0010	
the net rise or fall of the second-lowest voice is high (> 2) AND the melody's 7th note (pitch height) is high (> 68)	LIKE	+0.22	0.0023	
the melody's 9th note (pitch height) is high (> 70)	LIKE	+0.17	0.0037	

pattern	direction	effect size	p-value	verdict
the melody's 8th note (pitch height) is high (> 70)	LIKE	+0.16	0.0050	✓
the melody's 7th note (pitch height) is high (> 70)	LIKE	+0.14	0.0127	✓
average pitch of the melody (how high it sits) is high (> 70)	LIKE	+0.12	0.0383	✗ (looked significant on its own, but did not survive the correction for testing many patterns)
the net rise or fall of the second-lowest voice is high (> 0)	LIKE	+0.12	0.0430	✗ (looked significant on its own, but did not survive the correction for testing many patterns)
the melody's 6th note (pitch height) is high (> 70)	LIKE	+0.11	0.0457	✗ (looked significant on its own, but did not survive the correction for testing many patterns)
the net rise or fall of the second-lowest voice is low (≤ 2) AND the most consonant interval inside the final chord is low (≤ 1)	DISLIKE	-0.11	0.0536	✗
the net pitch change from the first chord to the last is high (> 0)	LIKE	+0.11	0.0600	✗
the melody's 9th note (pitch height) is low (≤ 72) AND the net rise or fall of the second-lowest voice is low (≤ 2)	DISLIKE	-0.11	0.0943	✗
the melody's 7th note (pitch height) is high (> 68) AND the net pitch change from the first chord to the last is high (> 2.25)	LIKE	+0.11	0.0953	✗
the melody's 8th note (pitch height) is high (> 68) AND the net	LIKE	+0.11	0.1040	✗

pattern	direction	effect size	p-value	verdict
pitch change from the first chord to the last is high (> 2.25)				
the net rise or fall of the third voice is high (> 0)	LIKE	+0.08	0.1629	✗
the most consonant interval inside the final chord is low (≤ 1) AND the net pitch change from the first chord to the last is low (≤ 2.25)	DISLIKE	-0.06	0.2836	✗

Patterns driven by the melody rather than the chords

These are the patterns whose deciding quantity comes from the melody line:

pattern	direction	effect size	p-value	verdict
the net rise or fall of the second-lowest voice is high (> 2) AND the melody's 6th note (pitch height) is high (> 68)	LIKE	+0.25	0.0007	✓
the melody's 8th note (pitch height) is high (> 68) AND the net rise or fall of the second-lowest voice is high (> 2)	LIKE	+0.23	0.0010	✓
the net rise or fall of the second-lowest voice is high (> 2) AND the melody's 7th note (pitch height) is high (> 68)	LIKE	+0.22	0.0023	✓
the melody's 9th note (pitch height) is high (> 70)	LIKE	+0.17	0.0037	✓
the melody's 8th note (pitch height) is high (> 70)	LIKE	+0.16	0.0050	✓
the melody's 7th note (pitch height) is high (> 70)	LIKE	+0.14	0.0127	✓
average pitch of the melody (how high it sits) is high (> 70)	LIKE	+0.12	0.0383	✗

pattern	direction	effect size	p-value	verdict
the melody's 6th note (pitch height) is high (> 70)	LIKE	+0.11	0.0457	✗
the melody's 9th note (pitch height) is low (≤ 72) AND the net rise or fall of the second-lowest voice is low (≤ 2)	DISLIKE	-0.11	0.0943	✗
the melody's 7th note (pitch height) is high (> 68) AND the net pitch change from the first chord to the last is high (> 2.25)	LIKE	+0.11	0.0953	✗
the melody's 8th note (pitch height) is high (> 68) AND the net pitch change from the first chord to the last is high (> 2.25)	LIKE	+0.11	0.1040	✗

How consistent were the ratings? (the ceiling on any result)

measure	value
average disagreement with the 5 most similar progressions	0.572
how often near-identical progressions got different ratings	0.675
accuracy from always guessing the most common rating	0.583
estimated best accuracy any method could reach	0.428